DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-400-03 — Replace VALVE, SWING CHECK 3" P300 (4"-P300-P-1008/A2)

Area	A400 Pumps	**Task**	VR Valve replacement in kind
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	P-102A — P-102A centrifugal pump — E 164'-0" N -18'-6" EL 100'-9"		
Equipment	P-102A	**Lines**	4"-P300-P-1008
Isometrics	ISO-4"-P300-P-1008 (generated); owner's drawings ISO-P-1008		
Permits	General work permit		

Reason

Discharge check valve slamming; replace in kind.

Scope of work

1. Isolate per the isolation list, drain and vent 4"-P300-P-1008 between the isolations.
2. Unbolt and remove VALVE, SWING CHECK 3" P300 (4"-P300-P-1008/A2) at E 164'-0" N -16'-6" EL 104'-1/8".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, SWING CHECK 3" P300 of the same size, class and face-to-face (CHECK3-P300), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 4"-P300-P-1008/A2, 4"-P300-P-1008/A1, 4"-P300-P-1008/A4

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1008	4"-P300-P-1008	4"-P300-P-1008	3	1 error(s), 4 warning(s) — see below

ISO-P-1008 (PDF): ../isometrics-imported/ISO-P-1008.pdf — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1008	1	error	sheet 1: CONT'D ON ISO-P-1009 prints E 164'-3 5/16" N -17'-6" EL 107'-6" for E 164'-3 5/16" N -16'-6" EL 107'-6": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 4"-P300-P-1008 meets 4"-P300-P-1009 (E 164'-3 5/16" N -16'-6" EL 107'-6")
ISO-P-1008	2	warning	weld 4"-P300-P-1008/W14 (E 134'-8 11/16" N 2'-0" EL 116'-6", sheet 2) has no label
ISO-P-1008	3	warning	W15 (4"-P300-P-1008/W15, sheet 3) is labelled again on sheet 2; it is 4.0 in from the nearest break
ISO-P-1008	2	warning	branch weld 3/4"-P300-SP-7024/W1 (E 136'-0" N 2'-0" EL 116'-9 5/16", sheet 2) has no label
ISO-P-1008	2	warning	branch weld 3/4"-P300-I-7008/W1 (E 135'-0" N 2'-0" EL 116'-9 5/16", sheet 2) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-102A	equipment	P-102A centrifugal pump		E 164'-0" N -18'-6" EL 100'-9"
4"-P300-P-1008/A2	item	VALVE, SWING CHECK 3" P300	4"-P300-P-1008	E 164'-0" N -16'-6" EL 104'-1/8"
4"-P300-P-1008/A1	item	GASKET 3" P300 COMPRESSED SHEET	4"-P300-P-1008	E 164'-0" N -16'-6" EL 103'-4-15/16"
4"-P300-P-1008/A4	item	GASKET 3" P300 COMPRESSED SHEET	4"-P300-P-1008	E 164'-0" N -16'-6" EL 104'-7-5/16"
P-102A-D	nozzle	Nozzle P-102A-D (discharge, 3" P300) on P-102A		E 164'-0" N -16'-6" EL 103'-4-7/8"
4"-P300-P-1008/A5	item	VALVE, GATE, WEDGE, OS&Y 3" P300	4"-P300-P-1008	E 164'-0" N -16'-6" EL 105'-2-3/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	P-102A-D	4"-P300-P-1008	Line end at nozzle P-102A-D: the equipment is out of service, drained and isolated under the shutdown isolation			
2	valve	4"-P300-P-1008/A5	4"-P300-P-1008	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (4"-P300-P-1008/A5)	E 164'-0" N -16'-6" EL 105'-2-3/16"		
3	blind	4"-P300-P-1008/A2	4"-P300-P-1008	Fit a line blind between 4"-P300-P-1008/A5 and VALVE, SWING CHECK 3" P300 (4"-P300-P-1008/A2), work side	E 164'-0" N -16'-6" EL 103'-5"		

Parts

Description	Qty	Replaces
VALVE, SWING CHECK 3" P300	1	4"-P300-P-1008/A2
GASKET 3" P300 COMPRESSED SHEET	1	4"-P300-P-1008/A1
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	4"-P300-P-1008/A3
GASKET 3" P300 COMPRESSED SHEET	1	4"-P300-P-1008/A4
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	4"-P300-P-1008/A6

Inspections

Id	Inspection	Criteria	Basis
WO-400-03-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
4"-P300-P-1008/A1	4 × 0.875" studs (4"-P300-P-1008/A3)	27,059	3	
4"-P300-P-1008/A4	4 × 0.875" studs (4"-P300-P-1008/A6)	27,059	3	

Test — Service leak test

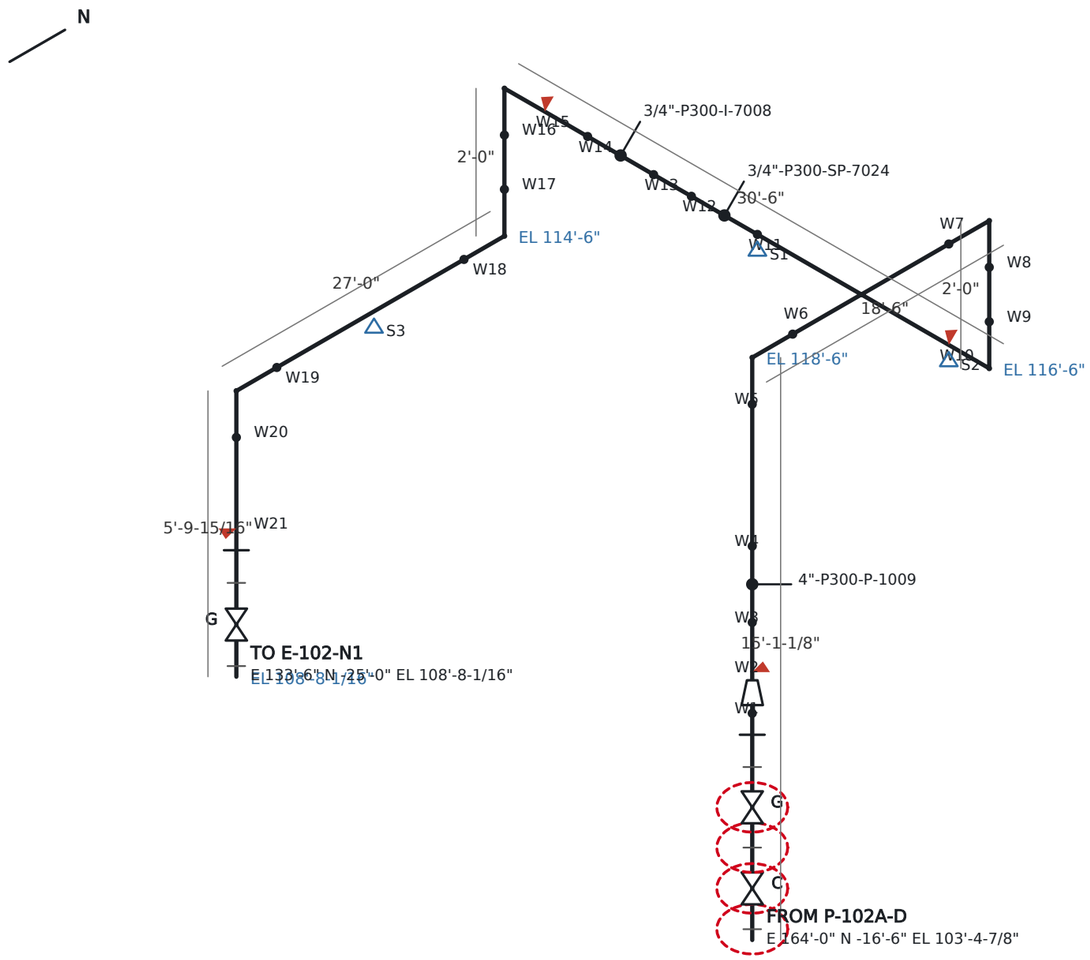
Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 4"-P300-P-1008/A2 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	CHECK3-P300	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P300-P-1008 — Replace VALVE, SWING CHECK 3" P300 (4"-P300-P-1008/A2)



BILL OF MATERIALS

1	GASKET 3" P300 COMPRESSED SHEET	3
2	VALVE, SWING CHECK 3" P300	1
3	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	3
4	VALVE, GATE, WEDGE, OS&Y 3" P300	1
5	FLANGE WELD-NECK 3" P300 RAISED FACE	1
6	REDUCER CONCENTRIC 4" × 3" P300	1
7	PIPE 4" P300 BEVELLED ENDS	87'-8-7/16"
8	TEE EQUAL 4" P300	1
9	ELBOW 90° R = 5/3 OD 4" P300	6
10	TEE REDUCING 4" × 3/4" P300	2
11	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	2
12	FLANGE WELD-NECK 4" P300 RAISED FACE	1
13	GASKET 4" P300 COMPRESSED SHEET	2
14	VALVE, GATE, WEDGE, OS&Y 4" P300	1

WELDS 21 (4 FIELD) · SUPPORTS 3 · SPOOLS 5

W1 W2F W3 W4 W5 W6 W7 W8 W9 W10F
W11 W12 W13 W14 W15F W16 W17 W18 W19 W20
W21F

4"-P300-P-1008

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
P-102A-D → E-102-N1
PIPE 87'-8-7/16" · WEIGHT 1192 lb

DWG ISO-4"-P300-P-1008 REV 0 WO-400-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Notices

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

This document embeds a subset of the DejaVu Sans font (SW-LIC-3), renamed under its licence. DejaVu Sans (assets/fonts/DejaVuSans.ttf), from the DejaVu fonts 2.37 as packaged in the build environment.

Copyright (c) 2003 by Bitstream, Inc. All Rights Reserved. Bitstream Vera is a trademark of Bitstream, Inc.
DejaVu changes are in public domain.

Permission is hereby granted, free of charge, to any person obtaining a copy
of the fonts accompanying this license ("Fonts") and associated
documentation files (the "Font Software"), to reproduce and distribute the
Font Software, including without limitation the rights to use, copy, merge,
publish, distribute, and/or sell copies of the Font Software, and to permit
persons to whom the Font Software is furnished to do so, subject to the
following conditions:

The above copyright and trademark notices and this permission notice shall
be included in all copies of one or more of the Font Software typefaces.

The Font Software may be modified, altered, or added to, and in particular
the designs of glyphs or characters in the Fonts may be modified and
additional glyphs or characters may be added to the Fonts, only if the fonts
are renamed to names not containing either the words "Bitstream" or the word
"Vera".

This License becomes null and void to the extent applicable to Fonts or Font
Software that has been modified and is distributed under the "Bitstream
Vera" names.

The Font Software may be sold as part of a larger software package but no
copy of one or more of the Font Software typefaces may be sold by itself.

THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS
OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF MERCHANTABILITY,
FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OF COPYRIGHT, PATENT,
TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL BITSTREAM OR THE GNOME
FOUNDATION BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, INCLUDING
ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL DAMAGES,
WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF
THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM OTHER DEALINGS IN THE
FONT SOFTWARE.

Except as contained in this notice, the names of Gnome, the Gnome
Foundation, and Bitstream Inc., shall not be used in advertising or
otherwise to promote the sale, use or other dealings in this Font Software
without prior written authorization from the Gnome Foundation or Bitstream
Inc., respectively. For further information, contact: fonts at gnome dot
org.